Bischoff 2021 LUAD — analysis summary

RUN 2 OF 3

GO / NO-GO GATES

Gate 1 — mixing & continuity → NO-GO

MIXED (diff-sample kNN frac 0.02→0.47, ceiling 0.81; both lobes internally sample-mixed, tumours span both) but NOT CONTINUOUS — two islands: PAGA between-lobe 0.01 vs within 0.55; 0.18% of edges & 1.9% of cells bridge; bimodal diffusion axis (peak/valley 387×). Invariant across θ {0,2,5,10}, n_pcs {12,16,20} — not under-integration.

Gate 2 — module geography → NA

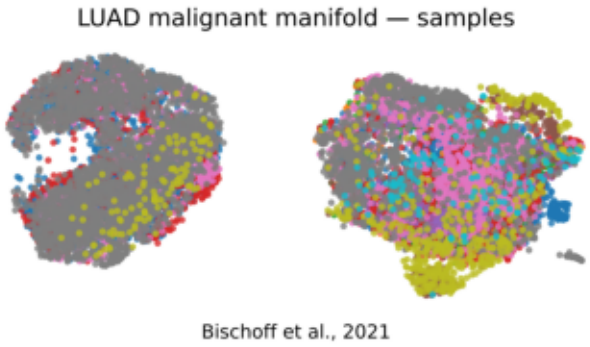
Not evaluated — Gate 1 NO-GO (spec §6). Lobe-identity fork: no lobe is a valid scope target — signatures near union-control baseline everywhere (Gast −0.04 / pSom +0.06 / Crest −0.03); the alveolar lobe shows the Kürten sign pattern but at baseline magnitude and is the DIFFERENTIATED pole (biologically backwards); the EMT/dediff lobe is incoherent (Gast~pSom −0.32).

Gate 3 — hub cross-repression → NA

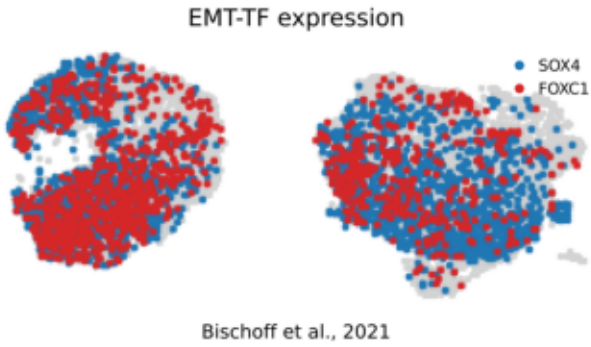
Not evaluated — branching-only, and no trajectory exists.

STEP FIGURES

Manifold — samples



EMT-TF expression



Dataset.
Lung adenocarcinoma (Bischoff et al., 2021), 10x — 13,382 malignant cells (10 tumour samples, 44-cell threshold).

Verdict.
Embryonic pattern does NOT reproduce — malignant structure is two disconnected EMT/differentiation lobes, not one continuous manifold; no trajectory fit (Gate 1 NO-GO). No lobe is a valid scope target: signatures sit at union-control baseline, and the one lobe with the Kürten sign pattern (alveolar) is the differentiated pole. Descriptive EMT-TF overlay (no trajectory to polarize): SOX4 graded high on the EMT/dediff lobe; SOX9/TEAD1/FOXC1 patchier.

Parameter choices

Min-cell threshold	44 (geom-mean of 127→15 gap, 8.5×) → 10/19 samples
n_pcs	16 (PCs > 0.5% variance; PC1 = 12.4%)
Embedding	k=15, UMAP spread=1.3, Leiden res=0.5 (12 clusters)
Signatures	score_genes vs union pool (Gast 80 / pSom 95 / Crest 103 genes)
Over-dispersed genes	1,976 (scFates GAM)
Integration	Harmony on sample, θ=2 (mixing plateaus by θ=2)
Lobe split	diffusion-axis valley: Alveolar 6,100 / EMT-dediff 7,282
EMT-TFs shown	SOX4 83%, SOX9 44%, TEAD1 44%, FOXC1 33% (≥3% floor; VIM excluded)